
SOUTHEAST ASIA ENGINEERING DESIGN COMPETITION 2026

BinSight

Smart waste sensing, recycling-return validation
and route simulation

Team: MON BLUE

Three sensor bins | QR return station | Local decision hub | Fixed-vs-priority route simulation

Engineering Proposal

Problem Statement

Collection crews still follow fixed routes, causing them to empty underfilled bins while high-use sites overflow. This wastes fuel, increases collection emissions, and leaves operators without real-time visibility into which bins need urgent service.

Proposed Solution

BinSight is a street-block prototype with three smart bins, one return station and hub. Focus A instruments each bin with a Teensy 4.1 running FreeRTOS for deterministic ultrasonic/load-cell polling, LED states, watchdog recovery and confidence flags for blocked or noisy readings. Focus C uses a Raspberry Pi/laptop hub to log sensor and QR events, estimate time-to-overflow with a tree-based model, rank pickups and compare fixed schedules against priority routes. Focus B adds a camera classifier that accepts plastic, metal and glass, then rejects non-recyclables before they enter the stream. Focus D builds an ESP32 return station following Singapore's Beverage Container Return Scheme / Return Right RVM flow with QR session, chute/servo feedback, dashboard logs and a simulated refund.

Simulation / Proposed Targets

The simulation uses prototype logs in a 30-day district model. It compares fixed collection with the priority route and checks for fewer overflows, fewer wasted trips, shorter route distance, lower fuel and CO2, cleaner return-station streams, and lower sensing energy within the USD150/SGD200 build.